Supplementary Material for The Epistemic Price of Coordination: Minimal Handshakes under Private Game Models

Anonymous Authors

1 Formal Model and Basic Properties

A finite two-agent epistemic coordination instance is

$$\mathcal{I} = (\Omega, X, Y, A, B, \tau_1, \tau_2, (u_\omega)_{\omega \in \Omega}),$$

where world ω determines a common payoff $u_\omega : A \times B \to \mathbb{R}$, while the sender and receiver observe only $x_\omega = \tau_1(\omega)$ and $y_\omega = \tau_2(\omega)$. An M-message deterministic protocol is $\pi = (\mu, \alpha, \beta)$ with $\mu : X \to [M]$, $\alpha : X \to A$, and $\beta : Y \times [M] \to B$. Let $v^\star_\omega = \max_{a,b} u_\omega(a, b)$ and

$$R(\pi) = \max_\omega \left[v^\star_\omega - u_\omega(\alpha(x_\omega), \beta(y_\omega, \mu(x_\omega)))\right].$$

Then $\mathrm{EPoC}_\mathcal{I}(M) = \min_\pi R(\pi)$ and $C_\mathcal{I}(\epsilon) = \min\{M : \mathrm{EPoC}_\mathcal{I}(M) \le \epsilon\}$, with the convention $\min \varnothing = \infty$.

Proposition 1 (Finite staircase and monotonicity)**.** *For every finite instance,* $\mathrm{EPoC}(M)$ *is attained, is nonincreasing in* M*, and belongs to the finite set*

$$\{v^\star_\omega - u_\omega(a, b) : \omega \in \Omega, a \in A, b \in B\}.$$

Moreover, $\mathrm{EPoC}(M) \le \epsilon$ *if and only if* $C(\epsilon) \le M$*.*

Proof. There are finitely many deterministic protocols, so a minimum is attained. Adding unused message symbols embeds every M-message protocol into the $(M + 1)$-message class. The regret of a fixed protocol is the maximum of finitely many world/action regret entries and hence equals one of them. The inverse equivalence follows directly from the definition of C. □

Proposition 2 (Zero-regret derandomization)**.** *A public-randomized M-message protocol has zero expected regret in every world if and only if a deterministic zero-regret M-message protocol exists.*

Proof. For each world, nonnegative regret has expectation zero only if it is zero almost surely over the public seed. Since the world set is finite, the intersection of these probability-one events also has probability one and is nonempty. Any seed in the intersection induces a deterministic protocol with zero regret simultaneously in every world. The converse uses a point mass on a deterministic protocol. □

For $\epsilon \ge 0$, write

$$\mathcal{R}^\epsilon_\omega = \{(a, b) : v^\star_\omega - u_\omega(a, b) \le \epsilon\}.$$

2 Zero-Communication Feasibility

Theorem 3 (CSP equivalence)**.** *A zero-communication protocol of regret at most ϵ exists if and only if the following CSP is feasible: one variable $A_x \in A$ for every $x \in X$, one variable $B_y \in B$ for every $y \in Y$, and the constraint $(A_{x_\omega}, B_{y_\omega}) \in \mathcal{R}^\epsilon_\omega$ for every world.*

Proof. A zero-communication protocol is exactly a pair of local decision rules $\alpha : X \to A$ and $\beta : Y \to B$. Setting $A_x = \alpha(x)$ and $B_y = \beta(y)$ satisfies every constraint exactly when the protocol meets the regret threshold in every world. The converse reads the local rules from a CSP assignment. □

Theorem 4 (Sharp action-cardinality boundary)**.** *If both agents have at most two actions, zero-communication feasibility is decidable in linear time after scanning the payoff tables. With three actions per agent, it is NP-complete even for $\epsilon = 0$ and payoffs in $\{0, 1\}$.*

Proof. For two actions, identify each local variable with a Boolean variable. For each world and each forbidden pair $(a, b) \notin \mathcal{R}_\omega^\epsilon$, add

$$(A_{x_\omega} \neq a) \vee (B_{y_\omega} \neq b).$$

This is a two-literal clause. An assignment satisfies all clauses if and only if it avoids every forbidden pair. There are at most four clauses per world, so implication-graph 2-SAT is linear in the encoded instance size.

For hardness with three actions, reduce graph 3-colorability. Given $G = (V, E)$, create sender type x_v and receiver type y_v for every vertex, and give both agents actions $\{1, 2, 3\}$. For each v, add an equality world of types (x_v, y_v) with payoff one exactly when the actions agree. For every edge $\{u, v\}$, orient it arbitrarily and add an inequality world of types (x_u, y_v) with payoff one exactly when the actions differ. A zero-regret protocol must have $\alpha(x_v) = \beta(y_v)$; call this common action the color of v. Every edge world enforces distinct endpoint colors. Thus a protocol induces a proper 3-coloring, and a proper coloring gives a protocol. Membership in NP is immediate. □

3 Compatibility Hypergraphs

Fix ϵ. A sender subset $S \subseteq X$ is compatible when there exist sender actions $(a_x)_{x \in S}$ and receiver responses $(b_y)_{y \in Y}$ such that $(a_x, b_{y_\omega}) \in \mathcal{R}_\omega^\epsilon$ for every world with $x_\omega \in S$. Let H_ϵ have vertex set X and hyperedges equal to inclusion-minimal incompatible subsets. We define $\chi(H_\epsilon) = \infty$ when a singleton hyperedge makes proper coloring impossible.

Theorem 5 (Compatibility coloring)**.** $C_{\mathcal{I}}(\epsilon) = \chi(H_\epsilon)$.

Proof. Let π be an M-message protocol. For every message m, its fiber $S_m = \mu^{-1}(m)$ is compatible, witnessed by the sender actions and receiver responses used by π on message m. Hence no hyperedge is monochromatic, so μ is a proper M-coloring.

Conversely, let $c : X \to [M]$ be a proper coloring. If one color class were incompatible, finiteness would give an inclusion-minimal incompatible subset inside the class, contradicting properness. Choose a compatibility witness for each class, use the color as message, and combine the witnesses into the local decision rules. The resulting protocol meets tolerance ϵ. □

Proposition 6 (Smallest higher-order obstruction)**.** *There is an instance with three sender types for which every pair can share a message but all three cannot; exactly two messages are required.*

Proof. Use one receiver type, a trivial sender action, receiver actions A, B, C, and acceptable sets $S_1 = \{A, B\}$, $S_2 = \{B, C\}$, and $S_3 = \{A, C\}$. Every pair intersects, while $S_1 \cap S_2 \cap S_3 = \varnothing$. Thus the pairwise conflict graph is empty, but the compatibility hypergraph has the single edge $\{1, 2, 3\}$. Two messages suffice by placing any two types together. □

Theorem 7 (Unbounded failure of pairwise confusability)**.** *For every prime-power projective-plane order q, there is a binary-payoff instance with $q^2 + q + 1$ sender types in which every pair is compatible, yet exact coordination requires exactly $q + 1$ messages.*

Proof. Use the lines of $\mathrm{PG}(2, q)$ as sender types, its points as receiver actions, one receiver type, and a trivial sender action. A line type receives payoff one exactly when the receiver selects a point on the line. Every pair of lines intersects, so each pair is compatible.

A message class is compatible exactly when its lines contain a common point. Equivalently, an M-message protocol selects at most M points hitting all lines. The $q + 1$ points of any fixed line hit every line, giving an upper bound. For a lower bound, take any set P of at most q selected points and choose a point $p \notin P$. There are $q + 1$ lines through p. Each selected point lies with p on exactly one of these lines, so at most $|P| \leq q$ of the lines meet P. At least one line through p is unhit. Therefore $q + 1$ messages are both necessary and sufficient. □

4 Specializations and Tractable Structure

Proposition 8 (Hitting-set specialization). *Suppose there is one receiver type, the sender action is trivial, rewards are binary, and sender type x accepts receiver actions $S_x \subseteq B$. Then*

$$C(0) = \min\{|P| : P \subseteq B,\ P \cap S_x \neq \varnothing\ \forall x\}.$$

Proof. A protocol chooses one receiver action per message. The selected actions hit the acceptable set of every type assigned to a message, so all selected actions form a hitting set. Conversely, given a hitting set, assign each sender type to a selected action in its acceptable set and use that action as the response to the corresponding message. □

Corollary 9 (Hardness and greedy approximation). *Minimum-message exact coordination is NP-hard in the setting of Proposition 8. Greedy maximum coverage uses at most $H_{|X|}C(0)$ messages.*

Proof. Proposition 8 is an approximation-preserving equivalence with hitting set, or dually set cover on the universe of sender types. NP-hardness and the standard harmonic-number charging analysis of greedy therefore apply. □

This is a classical zero-error coordination/set-cover connection rather than a new generic reduction; it is included to make the structural subclasses and experimental baselines explicit.

For a receiver-only instance with response space $B \subseteq \mathbb{R}^d$, let $S^{\epsilon}_{x,y}$ be the responses meeting tolerance ϵ in every world with type pair (x, y); for an impossible pair set $S^{\epsilon}_{x,y} = B$.

Theorem 10 (Helly-rank bound). *If every $S^{\epsilon}_{x,y}$ is convex, every inclusion-minimal incompatible set has size at most $d + 1$.*

Proof. If a sender class C is incompatible, then for some receiver type y, $\bigcap_{x\in C} S^{\epsilon}_{x,y} = \varnothing$. Helly's theorem gives a subfamily indexed by at most $d + 1$ sender types with empty intersection. Therefore every incompatible set contains an incompatible subset of size at most $d + 1$, and a minimal one cannot be larger. □

Corollary 11 (Intervals). *For one-dimensional interval response sets, pairwise compatibility implies joint compatibility. With one receiver type, minimum-message exact coordination is solved optimally by repeatedly selecting the right endpoint of the earliest-ending unhit interval.*

Proof. The first claim is Theorem 10 with $d = 1$. For the algorithm, let I be the remaining interval with smallest right endpoint r. Every feasible stabbing set contains some point $p \in I$. Replacing p by r cannot lose any remaining interval J previously hit by p: because $p \le r$ and r is the smallest right endpoint, $\ell(J) \le p \le r \le r(J)$. Hence an optimal solution containing r exists. Delete all intervals containing r and apply induction. □

5 Sparse Type-Interaction Graphs

Merge all worlds with the same type pair into the relation of action pairs that satisfy all of them. Form the bipartite type-interaction graph on $X \cup Y$, with edge (x, y) whenever that pair occurs. For fixed M, use sender variable $Z_x = (m, a) \in [M] \times A$ and receiver variable $W_y = (b_1, \dots, b_M) \in B^M$. The edge constraint accepts $((m, a), (b_1, \dots, b_M))$ exactly when (a, b_m) is acceptable for all worlds with pair (x, y).

Proposition 12 (Bounded-treewidth feasibility). *Given a width-w tree decomposition, fixed-M feasibility is decidable in*

$$O\big((|X| + |Y|)D^{w+1}\big), \qquad D = \max\{M|A|, |B|^M\},$$

up to polynomial local-constraint checks.

Proof. Run the standard dynamic program on a nice tree decomposition, recording the feasible assignments to each bag. Bags contain at most $w + 1$ variables, each with domain size at most D. Introduce, forget, and join transitions retain exactly the partial assignments satisfying all constraints whose endpoints have appeared. There are $O(|X| + |Y|)$ bags after standard normalization. □

6 Metric Action Dictionaries

Let (V, d) be a finite metric space with diameter D. Sender type x has a nonempty target uncertainty set U_x. Receiver type g is a bijective isometry from local action labels V to physical outcomes V, and payoff for local action b and target $t \in U_x$ is $1 - d(g(b), t)/D$. Define $\Delta(c, x) = \max_{t \in U_x} d(c, t)$.

Theorem 13 (Metric dictionary reduction)**.** *For every M,*

$$\mathrm{EPoC}(M) = \frac{1}{D} \min_{c_1,\dots,c_M,\, \mu: X \to [M]} \max_{x \in X} \Delta(c_{\mu(x)}, x).$$

Proof. Consider any protocol. For receiver type g and message m, its local response $b_{g,m}$ induces physical center $c_{g,m} = g(b_{g,m})$. The worst physical error on sender type x assigned to m is $\Delta(c_{g,m}, x)$. Since g is a bijection, every $c \in V$ is implementable as $g(g^{-1}(c))$. Furthermore, the family (U_x) is independent of g, so for each message class the minimax physical-center objective is identical for every receiver type. Replacing all $c_{g,m}$ by a common optimal c_m cannot increase regret. This maps every protocol to a partition and M centers with the displayed cost.

Conversely, given centers and a map μ, the sender transmits $\mu(x)$ and receiver type g chooses local action $g^{-1}(c_{\mu(x)})$. The physical outcome is the prescribed center, attaining exactly the displayed cost. □

When $U_x = \{x\}$, the objective is standard metric M-center. The classical farthest-first traversal therefore gives a 2-approximate deterministic handshake: select an arbitrary center, repeatedly add the point farthest from the current centers, and message the nearest selected center.

7 Closed-Form Additive Lever Frontier

Let $K = 2h+1$ levers form a cycle. The sender observes estimate x, while the true target is any t satisfying $d_K(x, t) \le r$. Receiver type y is a nominal action-frame shift, but the realized local-to-physical dictionary also contains a latent bias δ with $d_K(\delta, 0) \le s$. Local action b selects physical lever $b + y + \delta \pmod K$, and

$$u(t, b, y, \delta) = 1 - \frac{d_K(b + y + \delta, t)}{h}.$$

Lemma 14 (Odd-cycle covering radius)**.** *The minimum radius needed to cover an odd cycle of K vertices with at most M centers is*

$$\rho_K(M) = \left\lceil \frac{K - M}{2M} \right\rceil.$$

Proof. A radius-R ball contains at most $2R + 1$ vertices, so any cover requires $M(2R + 1) \ge K$, equivalent to $R \ge \lceil (K - M)/(2M) \rceil$. Conversely, if the inequality holds, partition the cyclic order into at most M consecutive arcs of size at most $2R + 1$ and choose a midpoint of each nonempty arc. □

Lemma 15 (Two-sided uncertainty dilation)**.** *For any estimate x and intended physical center c,*

$$\max_{\substack{d_K(t,x) \le r \\ d_K(\delta,0) \le s}} d_K(c + \delta, t) = \min\{h, d_K(c, x) + r + s\}.$$

Proof. The triangle inequality and cycle diameter give $d_K(c+\delta, t) \le d_K(c, x) + d_K(\delta, 0) + d_K(x, t) \le d_K(c, x) + s + r$ and also $d_K(c + \delta, t) \le h$. Let $d = d_K(c, x)$ and orient a shortest arc from x to c. If $d + r + s \le h$, move the realized center distance s farther in that direction and the target distance r in the opposite direction; their distance is $d + r + s$. If $d + r + s > h$, choose nonnegative $p \le r$ and $q \le s$ with $p + q = h - d$ (possible because $h - d < r + s$), move the target distance p opposite the arc and the center distance q along it, and obtain distance h. Thus the upper bound is attained in both cases. □

Theorem 16 (Exact additive private-lever EPoC)**.** *For every $1 \le M \le K$,*

$$\mathrm{EPoC}_{K,r,s}(M) = \frac{\min\left\{h, r + s + \left\lceil \frac{K-M}{2M} \right\rceil\right\}}{h}.$$

Proof. Fix a receiver frame y. Any deterministic M-message protocol induces at most M intended physical response centers, one for each used message. If estimate x is assigned to center c, Lemma 15 makes its worst physical error $\min\{h, d_K(c,x)+r+s\}$. Since this is nondecreasing in $d_K(c,x)$, minimizing the maximum error requires an optimal M-center cover of the estimates, whose radius is $\rho_K(M)$ by Lemma 14. This proves the lower bound for any fixed frame.

For the upper bound, choose an optimal set of cycle centers and send the index of the center assigned to x. Receiver type y implements intended center c with local action $c - y \pmod K$. The latent bias and target uncertainty then incur exactly the dilation in Lemma 15. The construction works for every observed nominal frame, so normalizing by h proves equality. □

Corollary 17 (Irreducible floor and inverse threshold)**.** *Full sender-type revelation leaves*

$$\mathrm{EPoC}_{K,r,s}(K) = \frac{\min\{h, r+s\}}{h}.$$

For $0 \le \epsilon < 1$*, define* $R = \lfloor \epsilon h \rfloor - r - s$*. If* $R < 0$*, no message budget attains the tolerance; otherwise*

$$C_{K,r,s}(\epsilon) = \left\lceil \frac{K}{2R+1} \right\rceil.$$

Proof. The floor follows by substituting $M = K$. For $\epsilon < 1$, the diameter cap is inactive at every feasible solution, and the threshold is equivalent to $\rho_K(M) \le R$, or $M(2R+1) \ge K$. □

8 Full-Revelation Floors and Ambiguous Action Dictionaries

Suppose the sender has one strategically trivial action. For each observed type pair, let

$$\Omega(x,y) = \{\omega : x_\omega = x,\ y_\omega = y\}, \qquad \delta(x,y) = \min_{b\in B} \max_{\omega\in\Omega(x,y)} [v^\star_\omega - u_\omega(b)].$$

Proposition 18 (Receiver-only full-revelation floor)**.** *If* $M \ge |X|$*, then*

$$\mathrm{EPoC}(M) = \max_{(x,y):\Omega(x,y)\neq\varnothing} \delta(x,y).$$

Proof. With $|X|$ messages, the sender can reveal its type x. The receiver then observes the cell (x,y) and can choose a minimax action attaining $\delta(x,y)$ independently for every nonempty cell, giving the upper bound. Conversely, even after exact revelation of x, every deterministic receiver response is one action for all latent worlds in the same cell. Its cellwise worst-case regret is therefore at least $\delta(x,y)$; taking the worst cell gives the lower bound. Extra messages cannot reveal which latent world inside a fixed (x,y) cell occurred because the sender observes only x. □

As a concrete corollary, let $K = 2h+1$. The sender observes the physical target x exactly. Receiver type y has a set $D_y \subseteq \mathbb{Z}_K$ of possible unobserved additive biases in its local-to-physical action map: local action b selects physical lever $b+y+\delta$ for some $\delta \in D_y$. Payoff is one minus normalized cyclic distance to x.

Corollary 19 (Full-revelation dictionary floor)**.** *If* $M \ge K$*, then*

$$\mathrm{EPoC}(K) = \frac{1}{h} \max_{y} \min_{c\in\mathbb{Z}_K} \max_{\delta\in D_y} d_K(c,\delta).$$

For $D_y = \{0, d_y\}$*, the inner value is* $\lceil d_K(0,d_y)/2 \rceil$.

Proof. With full target revelation, fix (x,y). Choosing local action b is equivalent to choosing an offset $c = x-(b+y)$; under bias δ, the physical error is $d_K(c,\delta)$. The cellwise minimax error is therefore $\mathrm{rad}(D_y) = \min_c \max_{\delta\in D_y} d_K(c,\delta)$, independent of x by translation invariance. Apply the preceding proposition and normalize by h. For two points on a cycle, a midpoint of a shortest arc achieves radius $\lceil d/2 \rceil$, and no smaller ball can contain endpoints at distance d. □

9 Robust Ambiguity Sets

Suppose each candidate model-generation process Q induces a possible-world set Ω_Q. For an ambiguity family $\mathcal{Q}$, define the robust instance by $\Omega_{\mathcal{Q}} = \bigcup_{Q \in \mathcal{Q}} \Omega_Q$.

Proposition 20 (Ambiguity monotonicity)**.** *If $\mathcal{Q} \subseteq \mathcal{Q}'$, then for every M and ϵ,*

$$\mathrm{EPoC}_{\mathcal{Q}}(M) \leq \mathrm{EPoC}_{\mathcal{Q}'}(M), \qquad C_{\mathcal{Q}}(\epsilon) \leq C_{\mathcal{Q}'}(\epsilon).$$

Proof. Every protocol's maximum regret can only increase when additional worlds are included. Taking the minimum over protocols proves the first inequality; the second follows from the inverse definition. □

10 MILP Formulation and Reproducibility Details

For fixed M and ϵ, introduce binary variables s_{xma} and r_{ymb}. Enforce

$$\sum_{m,a} s_{xma} = 1 \quad \forall x, \qquad \sum_{b} r_{ymb} = 1 \quad \forall y, m.$$

For every world ω, every unacceptable (a, b), and every message m, add

$$s_{x_\omega ma} + r_{y_\omega mb} \leq 1.$$

Feasibility is equivalent to an M-message protocol meeting the threshold. Searching the finite regret levels by binary search yields exact $\mathrm{EPoC}(M)$. The implementation decodes the protocol and independently reevaluates its worst-case regret. Only HiGHS status 2 is treated as a certificate of infeasibility. A resource limit without a feasible incumbent raises an explicit error rather than being silently used by the EPoC binary search.

The projective-plane experiments construct $\mathrm{PG}(2, q)$ over prime fields by normalizing nonzero vectors in $\mathbb{F}_q^3$ and using the incidence condition $\ell^\top p = 0$. The additive private-lever experiments enumerate all estimates, nominal receiver frames, targets, and latent dictionary biases. Besides the 25 plotted $K = 13$ configurations, an exhaustive audit checks all 169 combinations of r, s, and M for $K \in \{3, 5, 7\}$. The map experiment enumerates the eight square-grid isometries and cross-checks the general protocol MILP against exact finite metric center. Random set-system results use fixed seeds; interval exactness is cross-checked against the MILP before scaling the greedy algorithm. The repository test suite contains 29 unit, brute-force, and theorem cross-checks.